

Warehouse Slotting and Traffic Optimization

Progress meeting: approach, algorithms, and current evidence

ABC SLOTTING

AFFINITY

TRAFFIC-AWARE

GLOBAL OPTIMIZATION

Layout Management Project

5 August 2026

Progress at a glance

What is implemented

- ▶ ABC velocity classification and physical profile extraction
- ▶ Store-day affinity identification and affinity-aware placement
- ▶ Paper-replication C&TBSA clustering and shelf assignment
- ▶ Global, time-bounded CP-SAT placement optimization

Meeting focus

How the four methods fit together, what each optimizes, and how hard warehouse rules remain protected.

1,524

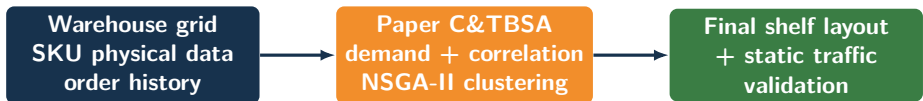
SKUs in velocity
summary

4

implemented stages

Source: Repository implementation and current generated artifacts.

C&TBSA is a direct pipeline



Not prerequisite stages

ABC SLOTting: COMPARISON

AFFINITY SLOTting: COMPARISON

GLOBAL CP-SAT: SEPARATE MODEL

Why a feasibility seed still exists internally

It reserves chilled and physical-exception storage and checks hard rules. Eligible standard-SKU membership is then discarded and rebuilt by C&TBSA.

Two constraint layers define the feasible C&TBSA layout

Paper model: reproduced

$$\sum_k x_{jk} = 1, \quad \sum_j x_{jk} \leq Z_k$$
$$\sum_j F_j x_{jk} \leq W_{\max}, \quad x_{jk} \in \{0, 1\}$$

Every SKU enters one cluster; each cluster fits its storage-location count; $W_{\max}$ is at least every cluster's demand.

Warehouse safety: added feasibility

- ▶ AMR shelf with one ordinary SKU per location
- ▶ unique occupied addresses
- ▶ chilled and ambient separation
- ▶ inherited attributes and physical capacity
- ▶ oversize, overweight, multi-slot, or incomplete racks fixed

C&TBSA searches only assignments satisfying both layers

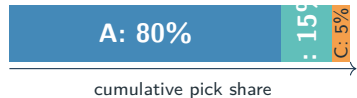
Source: Warehouse rules restrict feasibility; they do not change the paper's two objectives.

ABC identification: convert demand into velocity priority

$$f_i = \#\{\text{pick lines for SKU } i\}, \quad c_k = \frac{\sum_{i=1}^k f_{(i)}}{\sum_j f_j}$$

i, j : SKU indices; k : position in the sorted list; f_i : pick frequency;
 $f_{(i)}$: i th-highest frequency; c_k : cumulative pick share through rank k .

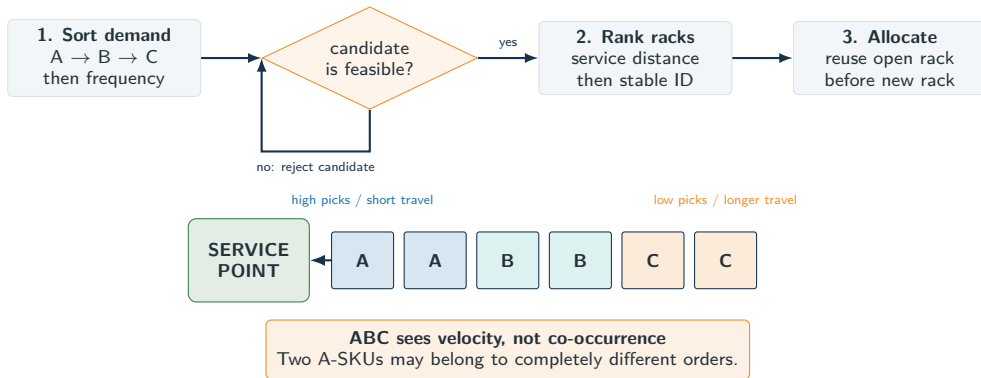
1. Count pick frequency per SKU.
2. Sort SKUs by descending f_i .
3. Compute cumulative demand share c_k .
4. Assign A through 80%, B through 95%, then C.



Why frequency, not quantity?

It targets handling events and travel triggers; unit quantity remains a separate inventory measure.

Pure ABC slotting algorithm



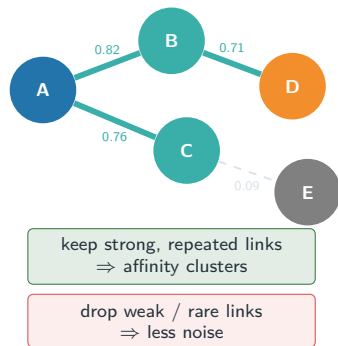
Affinity identification: find SKUs requested together

One fulfilment group is a **(Store ID, Date)** pair.
Presence is binary:

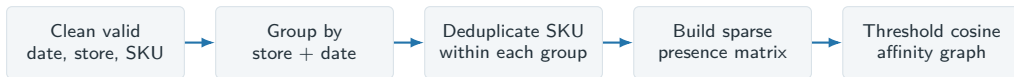
$$X_{g,i} = \begin{cases} 1, & \text{SKU } i \text{ occurs in group } g \\ 0, & \text{otherwise} \end{cases}$$

$$N(i,j) = \sum_g X_{g,i} X_{g,j}, \quad A(i,j) = \frac{N(i,j)}{\sqrt{N(i)N(j)}}$$

g : store-day group; i, j : SKUs; $N(i,j)$: shared-group count;
 $N(i) = \sum_g X_{g,i}$: groups containing SKU i ; $A(i,j)$: cosine affinity in $[0, 1]$.



Affinity identification workflow



Raw lines: duplicates retained for $F(i, s)$

	A	B	C
Store 1, Day 1	2	1	0
Store 1, Day 2	0	1	1

Binary presence used for affinity

	A	B	C
g_1	1	1	0
g_2	0	1	1

Duplicate A lines become one presence signal.

Source: Affinity analysis module and slotting support utilities.

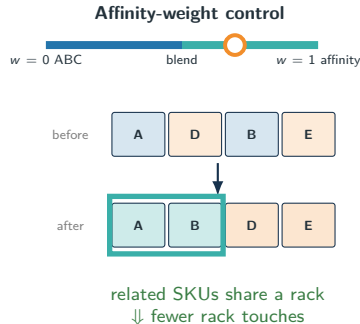
ABC + affinity slotting: blend two useful signals

Candidate ordering uses:

$$J_i = (1 - w) \underbrace{C_i}_{\text{ABC cost}} + w \underbrace{\left(1 - \frac{O_i}{O_{\max}}\right)}_{\text{affinity cost}}, \quad 0 \leq w \leq 1$$

J_i : selection cost; C_i : normalized ABC-class cost; w : affinity weight;
 O_i : candidate-to-cluster group overlap; $O_{\max}$: largest overlap among remaining candidates.

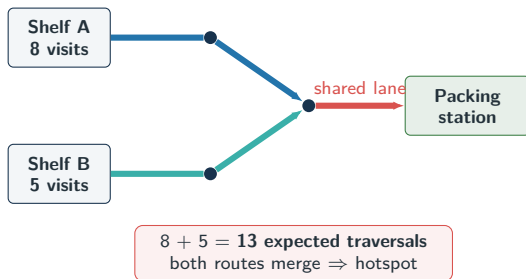
1. Start a rack cluster with the strongest demand signal.
2. Select the feasible SKU with minimum blended cost.
3. Update the cluster mask; reset when rack capacity is reached.



Notation reference: ABC and affinity

Symbol	Meaning
i, j	SKU indices; (i) denotes position after descending-frequency sorting.
k	Final rank included in a cumulative ABC calculation.
s, g	Store index s ; fulfilment-group index $g = (\text{Store ID}, \text{Date})$.
f_i, c_k	Pick frequency for SKU i ; cumulative pick share through rank k .
$F(i, s)$	Direct line-order frequency between SKU i and store s .
$X_{g,i}$	Binary indicator: 1 when group g contains SKU i , otherwise 0.
$N(i), N(i, j)$	Groups containing i ; groups containing both i and j .
$A(i, j)$	Cosine affinity between SKUs i and j , from 0 to 1.
J_i, C_i	Blended selection cost; normalized ABC-class component.
$w, O_i, O_{\max}$	Affinity weight; candidate overlap; maximum candidate overlap.
$\mathcal{F}, p$	Set of hard-rule-feasible placements; one candidate placement.

Why a good slotting layout can still create traffic



How load is calculated

1. Group orders by store and date.
2. Count visits to each complete shelf or tote.
3. Route every visit to a service endpoint.
4. Add the visits on every shared lane.

Optimization question

Can reassignment distribute expected demand while preserving affinity and physical feasibility?

Source: AMR counts one shelf visit per task; ASRS counts each required physical retrieval.

Updated traffic-aware workflow: Affinity first, then C&TBSA



What physically happens

1. Keep special physical inventory in its safe Affinity position.
2. Try many new groupings for ordinary standard SKUs.
3. Put the busiest selected groups on the nearest eligible shelves.

What it is trying to improve

Keep frequently co-requested SKUs together while preventing one shelf from receiving most of the demand. The model rebuilds only 1,498 standard SKUs; 26 exception outcomes remain unchanged.

Source: Implemented Traffic-Aware Slotting tab, Affinity-seeded C&TBSA workflow.

Affinity baseline: what is preserved and what C&TBSA changes

PRESERVED FROM AFFINITY

- ✓ same hard-rule allocator
- ✓ same chilled/ambient separation
- ✓ same oversize contiguous footprints
- ✓ same overweight required level
- ✓ same 26 exception outcomes



REPLACED BY C&TBSA

- 1,498 eligible standard SKUs
- affinity shelf grouping
- standard-SKU shelf selection
- standard-SKU position ordering
- selected Pareto trade-off

Meaning of “same approach”

Physical inventory is treated exactly as in Affinity. The complete final layout is intentionally different because C&TBSA must re-cluster standard SKUs to optimize its two paper objectives.

Step 1: turn order history into two simple scores

Small example first

Store + day group	SKU 17	SKU 04
Store A, Monday	✓	✓
Store B, Monday	✓	-

SKU 17 is requested in **2** groups.

SKU 04 is requested in **1** group.

They appear together in **1** group.

Plain-language question

How popular is each SKU, and how often are two SKUs requested together on the same store delivery day?

Mathematical proof used by the code

$$F_j = \sum_g X_{g,j}, \quad N_{jj'} = \sum_g X_{g,j} X_{g,j'}$$

g = one Store ID + Date group. j, j' = SKU indices.
 $X_{g,j} = 1$ if group g requests SKU j ; otherwise 0.
Therefore: $F_{17} = 2$, $F_{04} = 1$, and $N_{17,04} = 1$.

982,992 LINES → 12,398 GROUPS

1,524 SKUs → 1,498 OPTIMIZED

Step 2: describe one possible shelf plan

Write every eligible SKU once, shuffle the list, then cut it into shelf-sized groups.

Shelf group $k = 1$

17	04	31	09	22	11	D
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Shelf group $k = 2$

06	29	15	03	27	18	D
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No duplication

Every SKU appears in exactly one group: $\sum_k x_{jk} = 1$.

No overfilling

Shelf k holds at most Z_k locations:
$$\sum_j x_{jk} \leq Z_k.$$

$x_{jk} = 1$ means SKU j is assigned to shelf group k ; otherwise 0.

Unused space

D is a zero-demand dummy gene representing an empty shelf location.

Step 3: score every plan on two business goals

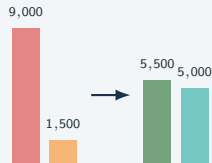
Goal A: keep requested-together SKUs together

SKU 17 + SKU 04 → SAME SHELF

$$\max f_1 = \sum_{j < j'} \sum_k N_{jj'} x_{jk} x_{j'k}$$

Larger f_1 means more frequently co-requested SKU pairs are stored on the same shelf.

Goal B: avoid one shelf carrying most demand



$$W_k = \sum_j F_j x_{jk}, \quad \min f_2 = W_{\max}$$

W_k is demand assigned to shelf k ; $W_{\max}$ is the busiest shelf.
Smaller $W_{\max}$ means a more even workload.

Important boundary

These objectives balance **SKU grouping and shelf workload**. They do not directly minimize AMR trips, travel distance, queues, or shared-lane congestion.

Step 4: let the computer improve many plans repeatedly



Technical names, translated

NSGA-II
PMX

repeats the four-step search above
combines two parent SKU orders without duplicating SKUs

2-opt swap
Pareto set

exchanges two positions to create a new plan
plans where neither goal can improve freely

Paper parameters

$$P = 100, G = 50,000$$

$$P_c = 0.9, P_m = 0.1$$

Current reproducible run

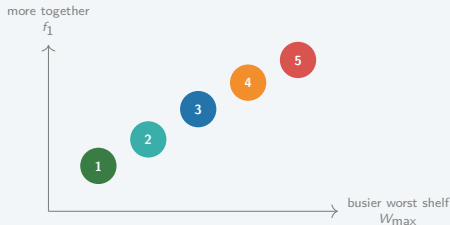
$$P = 100, G = 500$$

$$P_c = 0.9, P_m = 0.1, \text{seed} = 0$$

P = number of plans; G = repeated generations; P_c = PMX chance; P_m = swap chance.

Step 5: choose a trade-off and place busy groups closer

No single plan wins both goals



Five representative Pareto solutions are shown. The saved run selects **solution 5**, favouring co-request grouping.

Then map groups to real shelves

1. Calculate each group workload W_k .
2. Rank groups from busiest to quietest.
3. Rank eligible shelves from nearest to farthest.
4. Match both rankings; randomize positions inside each shelf.

BUSIEST GROUP



NEAREST ELIGIBLE SHELF

Exactly what is reproduced, adapted, and not claimed

	Implemented meaning	Current status
Paper method	permutation chromosome; two objectives; NSGA-II; PMX; swap mutation; Pareto representatives; demand-ranked Stage 2	reproduced after the Affinity baseline; the paper search itself is unchanged
Warehouse adaptation	AMR shelf as storage area; 12 locations; ambient/chilled strata; fixed physical exceptions; dummy empty genes	feasibility added without inserting lane load into the paper objectives
Static validation	Store ID + Date shelf visits routed over the saved grid to service endpoints	heatmaps, peak, P95, expected travel, and relocation report
Not reproduced yet	fleet dispatch, queues, blocking, collisions, picking delay, throughput	no simulation validation; no operational throughput claim

Interpretation boundary

A lower $W_{\max}$ proves a more balanced assignment-demand objective. It does not prove that total lane traffic or travel decreased.

Source: Lee, Chung, and Yoon, C&TBSA, DOI: 10.1016/j.cie.2019.106129.

Implementation proof: the traffic-balance objective is active

What the optimizer calculates

$$W_k = \sum_j F_j x_{jk}$$

$$\boxed{\min W_{\max}}, \quad W_{\max} = \max_k W_k$$

W_k is expected pickup demand assigned to shelf cluster k . Lower $W_{\max}$ means less demand concentrated in the busiest cluster.

How to read the evidence

Moving from solution 1 to solution 5 increases correlation, but also increases the maximum cluster demand. This is the intended Pareto trade-off from the paper.

Source: Values read from the ambient and chilled Pareto rows in the saved Traffic-Aware layout.

Saved run: five actual Pareto representatives

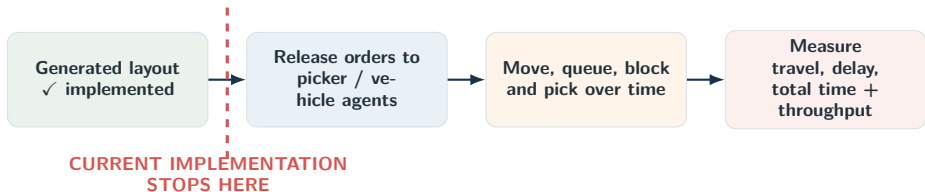
Sol.	Ambient		Chilled	
	$W_{\max}$	Correlation	$W_{\max}$	Correlation
1	9,745	419,521	5,852	31,335
2	10,467	432,044	7,378	39,540
3	11,351	479,335	8,701	49,397
4	12,031	499,629	10,547	58,838
5	14,096	520,396	14,188	66,008

SOLUTION 1: MORE TRAFFIC BALANCE →

SOLUTION 5: MORE CORRELATION

Current saved layout uses **solution 5**; solution 3 is the clearer middle trade-off for the full demonstration.

Paper component not implemented: operational simulation



Missing operational behaviour

- ▶ picker / AMR dispatch and order release timing;
- ▶ aisle occupancy, queues, blocking and conflicts;
- ▶ movement speed, picking/service time and station waiting;
- ▶ repeated stochastic runs with different random seeds.

What cannot yet be claimed

We can prove a lower cluster-demand proxy $W_{\max}$. We cannot yet prove lower congestion delay, faster order completion, or higher throughput. The current heatmap is static expected flow, not a time-dependent simulation.

Development needed for a complete Traffic-Aware demonstration

1. Add a discrete-event simulation engine

event clock; picker/AMR agents; route occupancy; queues; blocking; picking and workstation service events

2. Add inputs and UI controls

vehicle count; speed; pick/service time; aisle direction/capacity; order release rule; replications and random seed

3. Add a controlled experiment runner

freeze one dataset; run Basic, Affinity and C&TBSA solutions 1–5; use identical orders, hard rules and simulation seeds

4. Add operational result exports

travel time; congestion delay; total completion time; throughput; P95 queue; confidence intervals and animated/temporal heatmaps

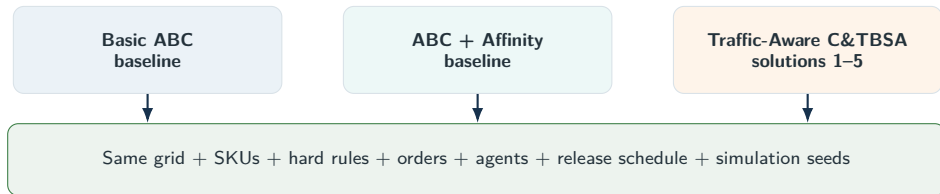
Code areas to extend

`traffic.py`: simulation adapter and experiment metrics `gui.py`: controls, run/cancel and result views
`new traffic_simulation.py`: event engine `new traffic_experiment.py`: repeatable multi-layout comparison

Keep the paper method intact

Do not insert lane congestion into the C&TBSA objective when claiming paper replication. Use simulation to validate each Pareto layout, as the paper does.

Full demonstration: compare three layout-generation methods fairly



Optimization-level evidence

- ▶ maximum cluster demand $W_{\max}$;
- ▶ co-request correlation f_1 ;
- ▶ shelves per order and occupied shelves;
- ▶ static travel and route-load heatmaps.

Operational simulation evidence

- ▶ mean and P95 travel time;
- ▶ mean and P95 congestion delay;
- ▶ total order time $T = T_{\text{travel}} + T_{\text{pick}} + T_{\text{delay}}$;
- ▶ throughput and 95% confidence interval.

Decision rule

Select the layout that improves total operational time without unacceptable travel, congestion, relocation, or hard-rule violations—not simply the layout with the lowest $W_{\max}$.

Notation reference: C&TBSA and global model

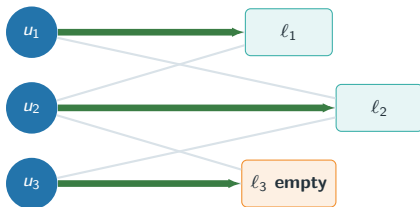
Symbol	Meaning
g	Fulfilment group: one unique (Store ID, Date) pair.
j, j'	Two different SKU indices; $j < j'$ counts each SKU pair once.
k	C&TBSA cluster index; mapped here to one eligible AMR shelf.
$X_{g,j}$	Binary presence: 1 when group g requests SKU j ; otherwise 0.
F_j	Demand for SKU j : number of selected groups containing it.
$N_{jj'}$	Number of selected groups containing both SKUs j and j' .
x_{jk}	Binary assignment: 1 if SKU j belongs to cluster k ; otherwise 0.
Z_k	Storage locations available in cluster/shelf k ; here at most 12.
$W_k, W_{\max}$	Assigned cluster demand; maximum demand across all clusters.
f_1, f_2	Correlation-maximization and maximum-demand-minimization objectives.
P, G, P_c, P_m	Population, generations, PMX probability, and swap-mutation probability.
u_i, ℓ_r	Complete handling unit i and candidate location r in global CP-SAT.

Global approach: assign all units together

Decision: Which compatible location should **every complete unit** occupy in the final layout?

COMPLETE UNITS

ALLOWED LOCATIONS



grey = feasible option
green = selected assignment

ONE UNIT PER LOCATION

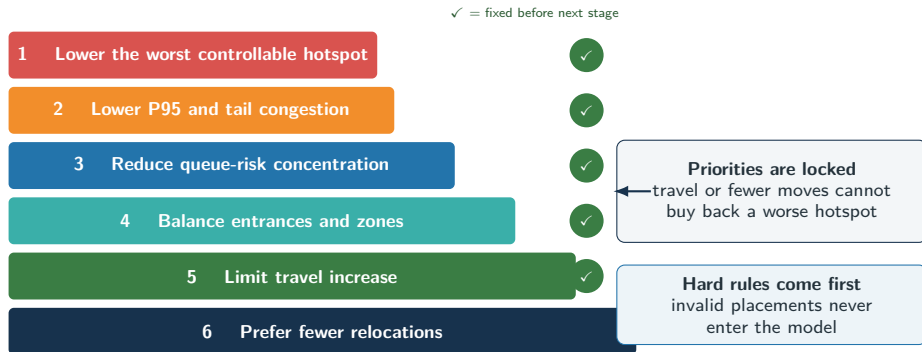
WHOLE LAYOUT EVALUATED

CP-SAT model

1. Remove locations that fail hard rules.
2. Assign each unit once and use each location at most once.
3. Calculate traffic for the whole assignment.
4. Keep the best solution found within the time limit.

Enables: empty-buffer moves, pair swaps, and multi-unit cycles in one model.

How the global optimizer chooses the winner



The solver records **OPTIMAL** only with proof. A time-limited valid result is **FEASIBLE**, with its bound or gap reported.

Traffic-aware or global: which one should we use?

TRAFFIC-AWARE: C&TBSA



Balance demand and preserve within-cluster affinity

PARETO FRONT PAPER REPLICATION

GLOBAL: JOINT



Choose every final location and score the complete layout

CYCLES EMPTY BUFFERS PROOF STATUS

Choose according to the research question

C&TBSA

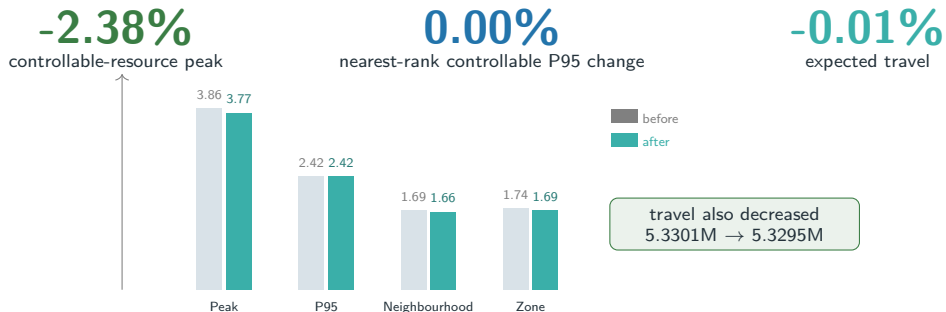
Reproduce correlation + workload balance

Global CP-SAT

Optimize mapped lane loads

C&TBSA follows the paper's two-stage method. Global CP-SAT is our separate warehouse-specific model; its objective directly uses the grid network.

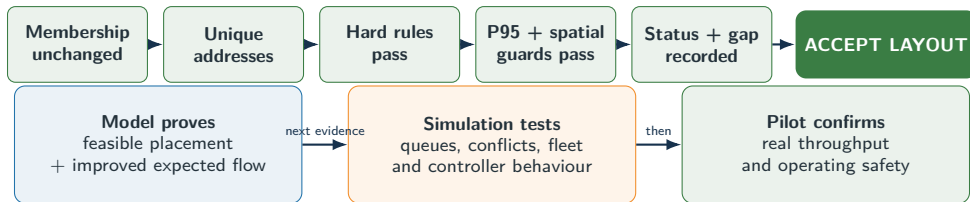
Current evidence: saved global run



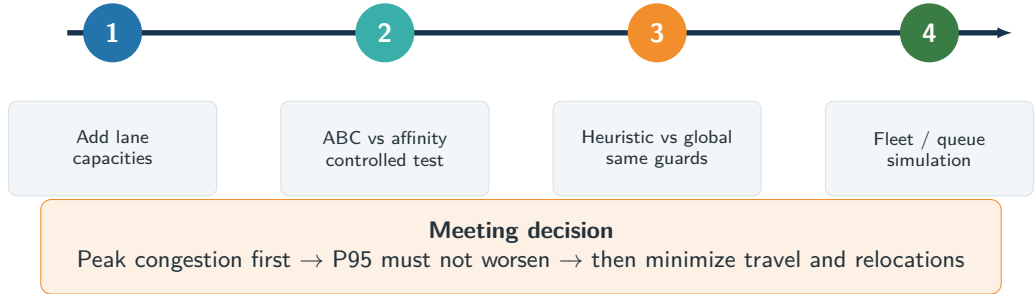
128 handling units, 158 candidate locations, 7,917 binary variables, 31 relocated units; 180 s result status: **FEASIBLE**.

Source: resources/map/global_traffic_optimized_layout.slotting.json. Relative loads are not physical utilization because the network has no explicit capacities.

Validation logic and interpretation boundary

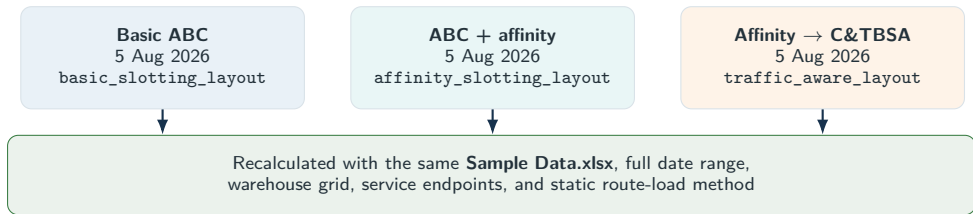


Next progress milestone



Implemented approach → controlled comparative validation

Comparison protocol: three saved layouts, one measurement method



Controlled static comparison

All three layouts now use the same canonical grid, SKU/chilled inputs, order workbook, hard-rule profile, and route calculation. Each assigns 1,523 SKUs and excludes the same dimension-incompatible SKU 139112. This is static expected-flow evidence, not simulation or throughput proof.

Assignment coverage and warehouse-space use

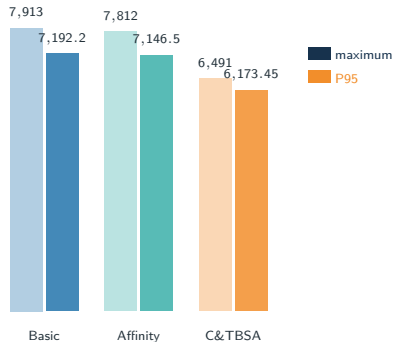
Metric	Basic ABC	Affinity	C&TBSA S5
Input SKUs	1,524	1,524	1,524
Assigned SKUs	1,523	1,523	1,523
Unassigned SKUs	1	1	1
Occupied shelves	132	132	158
Mean SKUs / occupied shelf	11.54	11.54	9.64
ABC-pure shelf fraction	95.5%	74.2%	5.7%
Mixed-ABC shelves	6	34	149

Basic packs inventory
into the fewest shelves

Affinity accepts more mixing
to consolidate orders

C&TBSA uses all shelves
to distribute demand

C&TBSA succeeds at its shelf-demand balancing objective



Measured shelf visits

	Maximum	CV
Basic	7,913	0.700
Affinity	7,812	0.708
C&TBSA	6,491	0.537

Result versus basic

18.0% lower maximum

23.3% lower CV

$CV = \sigma_v / \bar{v}$; v is expected visits to one shelf, $\bar{v}$ its mean, and σ_v its standard deviation. Lower is more even.

But shelf balance increased total retrieval and route load

Metric (lower is better)	Basic ABC	Affinity	C&TBSA S5
Total shelf visits	441,862	432,668	547,519
Average shelves / group	35.64	34.90	44.17
P95 shelves / group	77	76	101
Expected travel	5.234M	5.282M	7.229M
Raw peak route load	88,372.4	86,533.6	109,503.8
Weighted related-SKU distance	5.39 m	4.48 m	6.90 m
Related-pair same-shelf rate	2.48%	2.58%	1.04%

C&TBSA gain
more even shelf workloads

current trade-off

observed cost
more shelves per order + longer routes

Source: Static expected-flow recalculation; the grid has no explicit resource capacities, so raw load is not physical utilization.

Meeting conclusion: method works, operational trade-off is unresolved

Basic ABC

strongest ABC purity
fewest occupied shelves
simple baseline

Affinity

fewest total shelf visits
shortest related-SKU distance
lowest observed route peak

C&TBSA solution 5

lowest maximum shelf demand
most even workload
highest retrieval + travel cost

Evidence-based interpretation

The algorithm reproduction is functioning: its demand-balancing objective improved. The current selected Pareto solution does **not** yet provide the desired combined outcome of balanced shelves and low travel.

Next controlled experiment

Compare C&TBSA solutions 1–5 and seeds 0–2 on this frozen input snapshot, then validate the shortlisted trade-off in simulation.